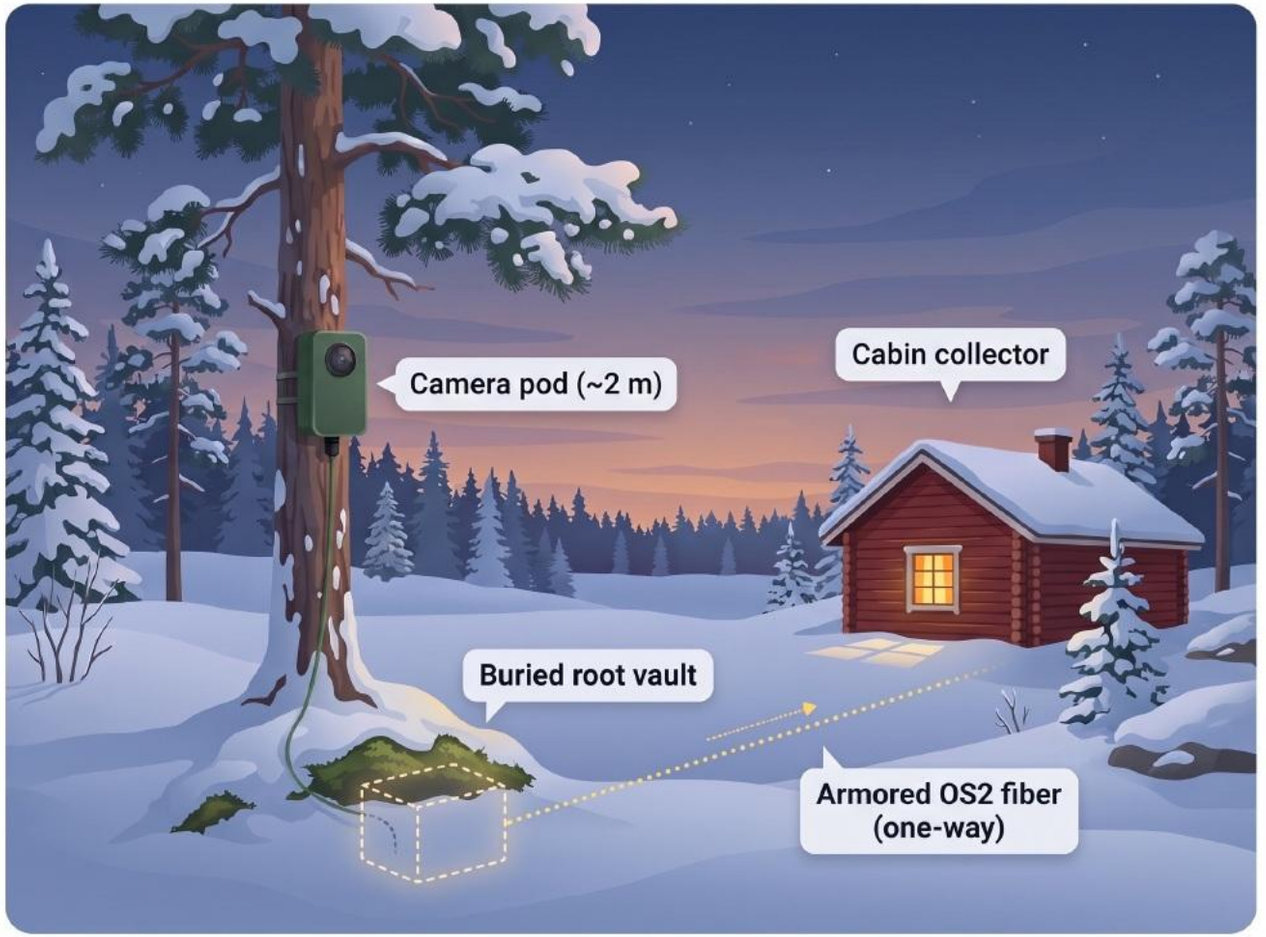
Arctic Trail Cam & Message Relay System

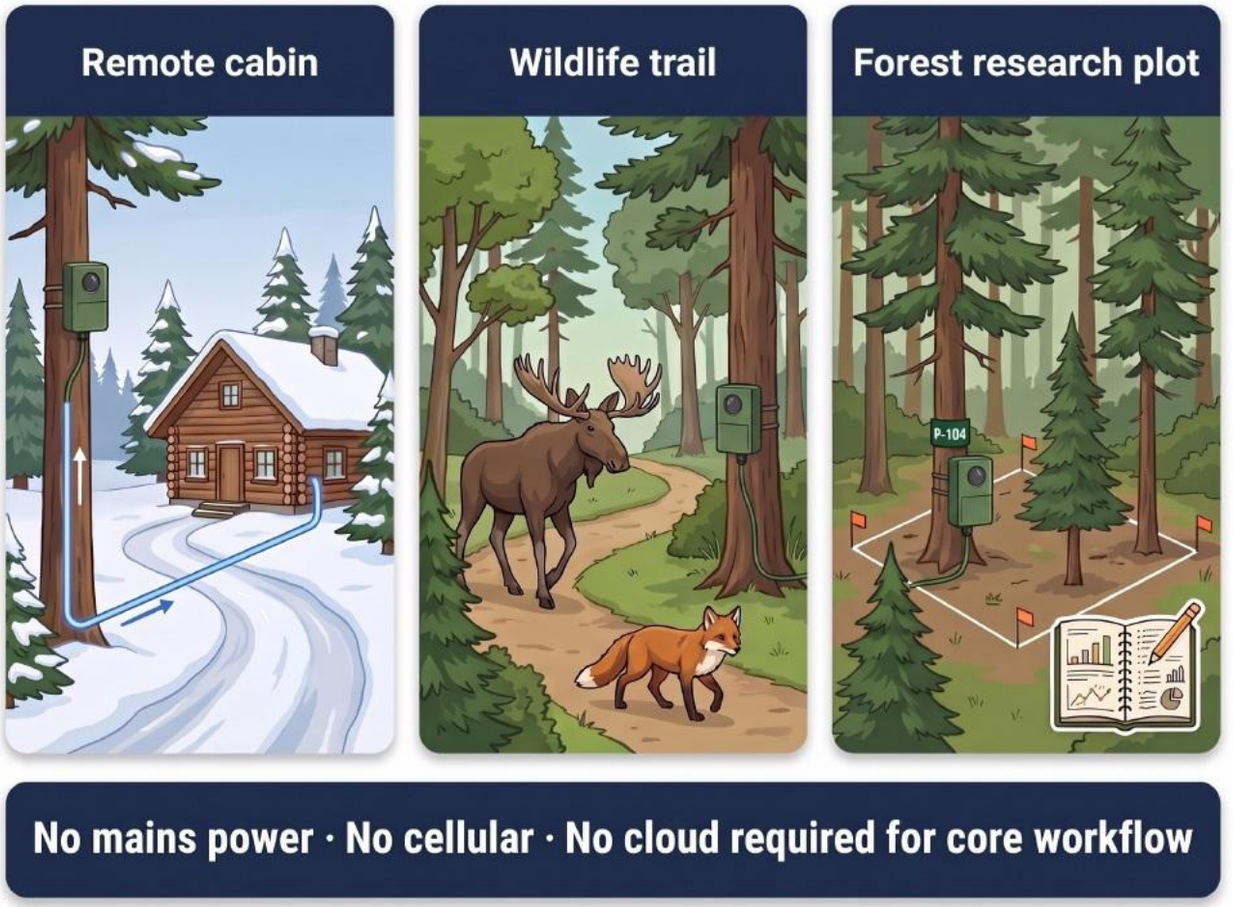
Off-grid detection, signed imaging and one-way delivery for remote sites

v0.9-beta experimental prototype · 20 Oct 2026

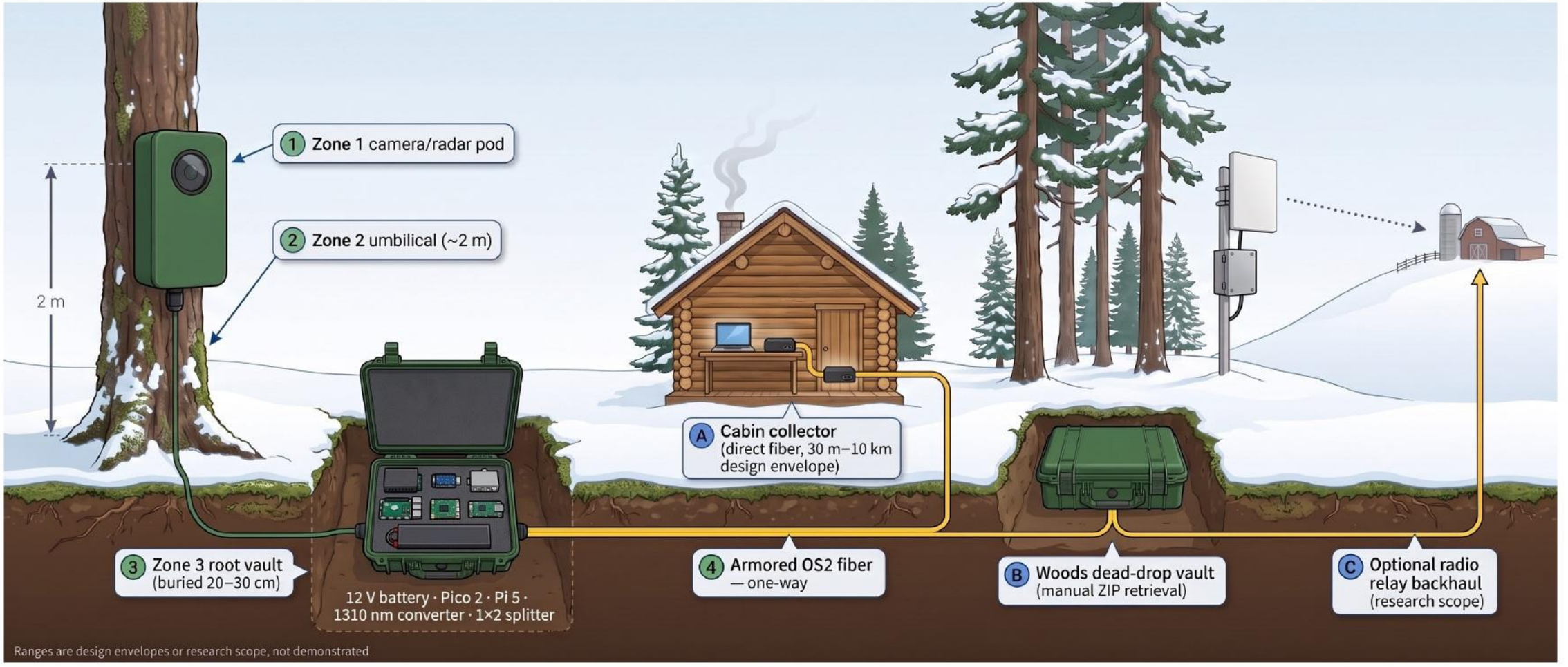


Purpose: Evidence-Quality Images Off-Grid

- **Who it serves:** remote cabin owners, wildlife-trail observers and forest research plots in eastern and northern Finland
- **What it does:** wakes on a qualifying event, takes a 1080p photo, analyses it locally, signs it and sends it **one way** to a local collector
- **Design priorities:** very low standby power, small visual footprint, no return path into the field node, no SIM card or subscription
- **Status:** v0.9-beta. Software and Stage 1–3 bench bring-up verified. **Field performance not yet demonstrated**



Complete Landscape Architecture: Sense at the Tree, Decide at the Roots, Deliver One Way



Three-Zone Tree Cutaway: A Small Eye Up High, the Heavy Parts Buried Below

ZONE 1 — Camera/radar pod, ~2 m up

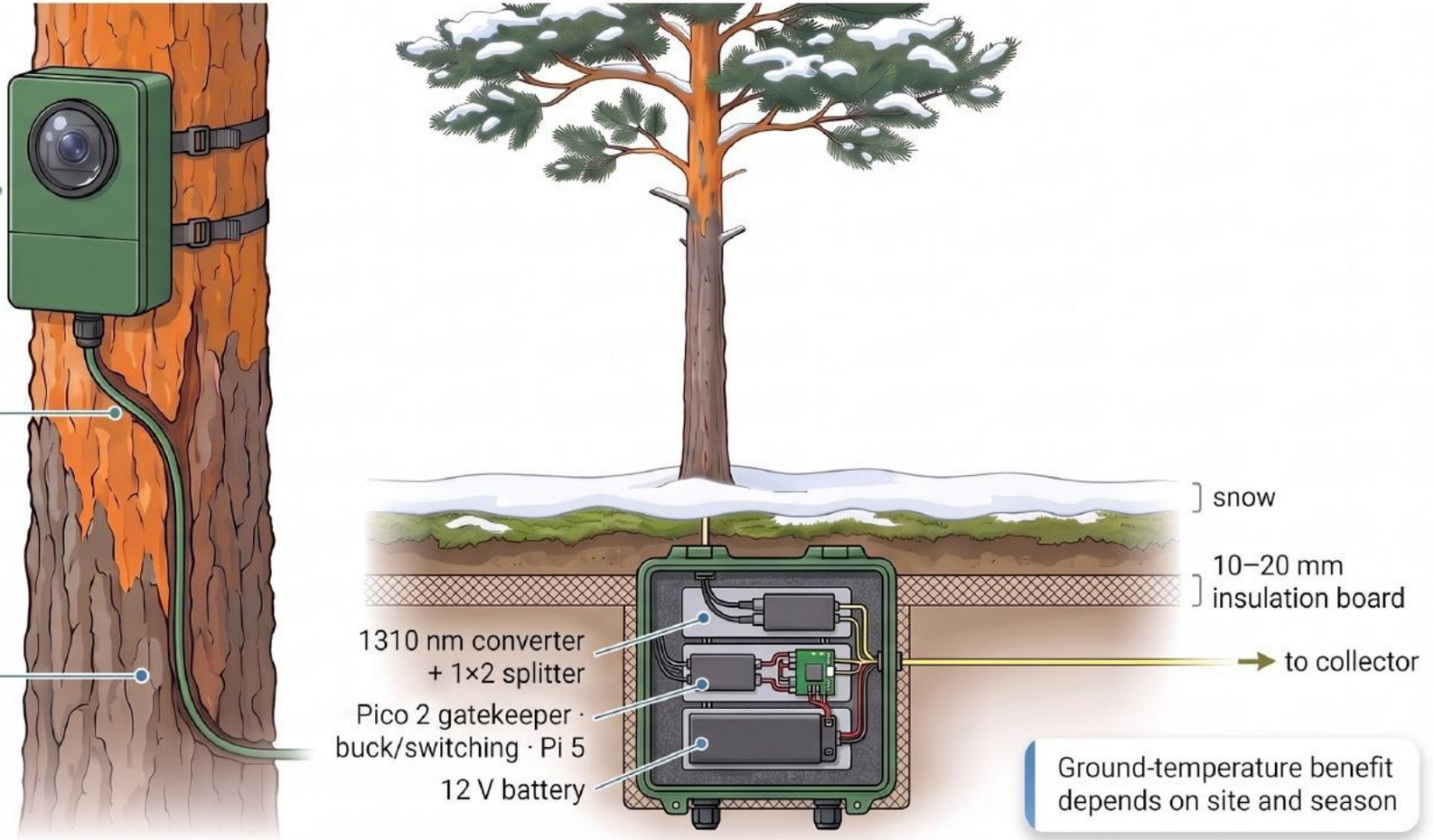
Hammond 1591BSBK enclosure

ZONE 2

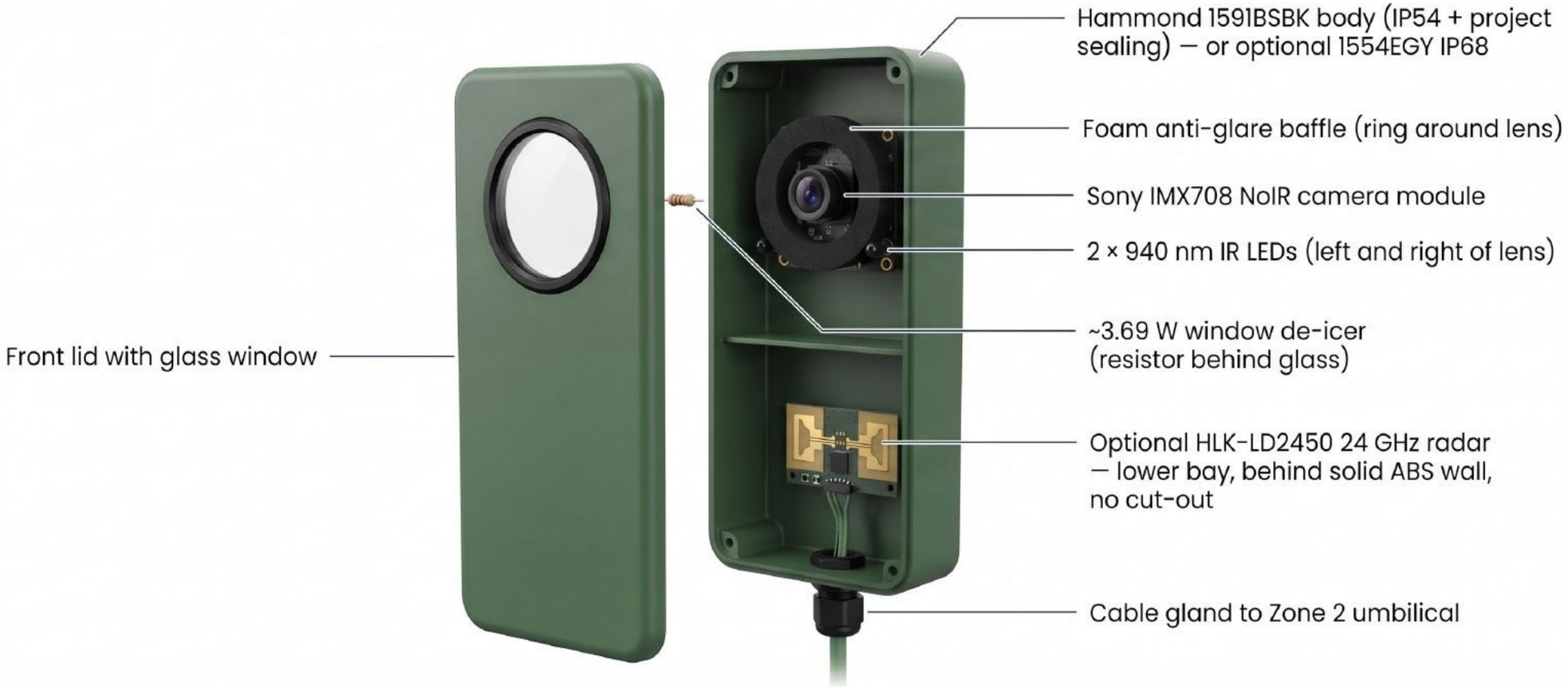
~2 m shielded CSI-to-HDMI extension + radar/heater umbilical

ZONE 3

QILIPSU root vault (nominal IP67), buried 20–30 cm

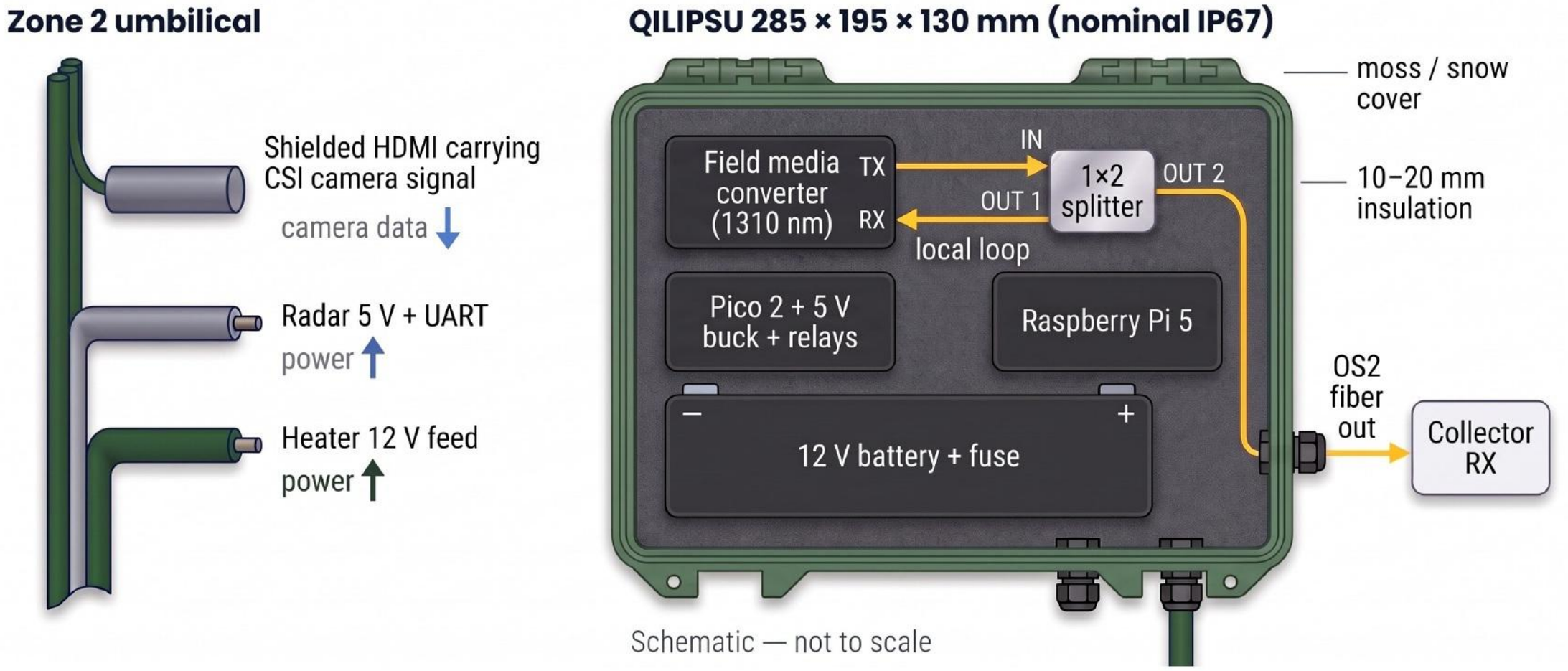


Camera Pod Exploded View: One Window, One Camera, Hidden Radar



Modified assembly has no independently established IP rating

Umbilical and Root Vault: Power Goes Up, Camera Data Comes Down



Sensor Roles: Three Different Physical Signals, Each with Blind Spots

QMC5883L magnetometer



Magnetic disturbance sensing

Senses magnetic/ferrous disturbance.
Does NOT reliably detect animals or people without metal.

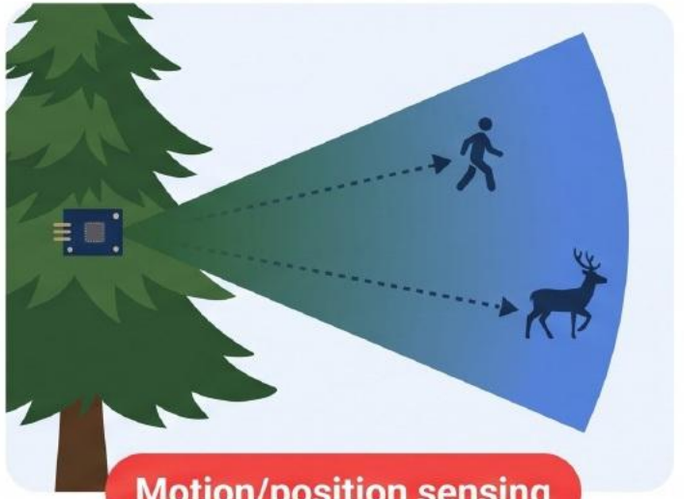
SM-24 geophone
+ ADS1115 (optional)



Ground vibration sensing

Senses ground vibration.
Sensitive to soil, frost and wind noise.

HLK-LD2450
24 GHz **radar** (optional)

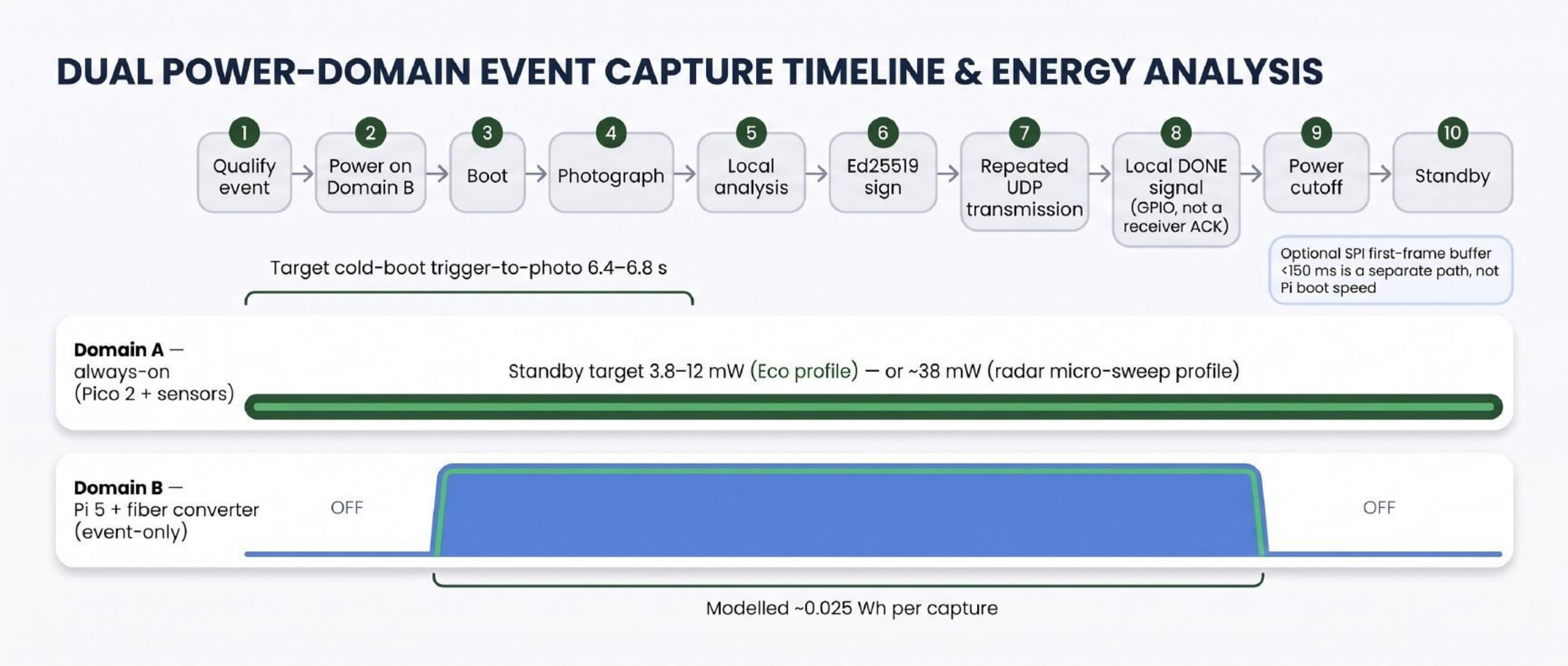


Motion/position sensing

Senses motion and position.
Active RF emitter when enabled.

Pico 2 gatekeeper polls sensors according to profile

Wake and Power Sequence: The Pi 5 Is Off Until an Event Qualifies



Capture and Local AI: Detection Runs at the Tree, Not in a Cloud

Stage 1: Picamera2 capture — fixed hyperfocal focus



Sensor Input & Initial Processing

Stage 2: Detection — three-tier fallback

Hailo-8L accelerator (optional)

Optimal Performance.
AI Acceleration.

↓

if absent, CPU YOLOv8n ONNX

Secondary Path.
Standard Model.

↓

if unavailable, Motion-ROI fallback

Tertiary Path.
Basic Motion Detection.

Stage 3: Output (illustrative example)



Example metadata (illustrative)

```
label: moose
confidence: 0.87
box: x, y, w, h
```

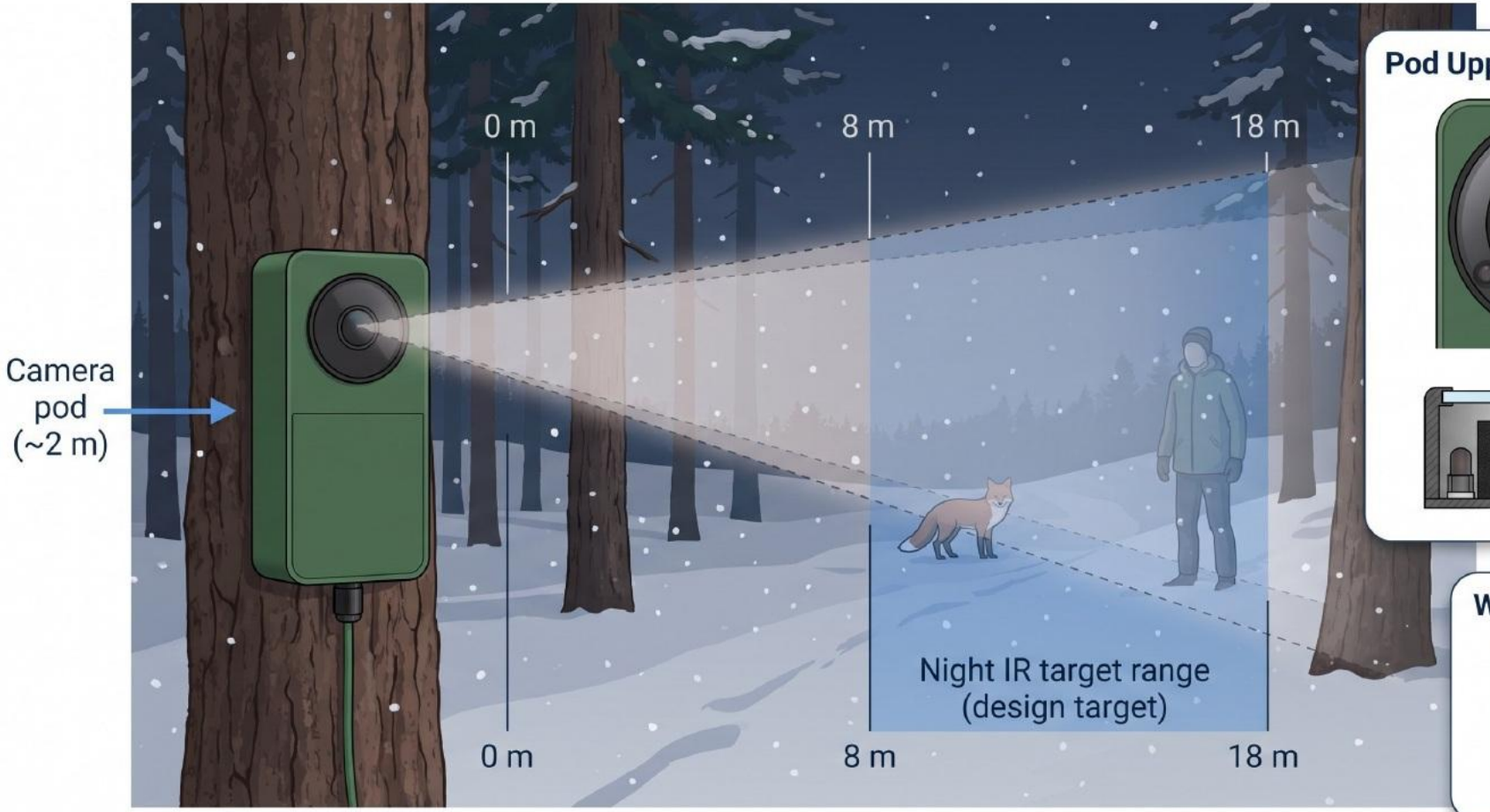
Stage 4: Ed25519 signing → one-way fiber send



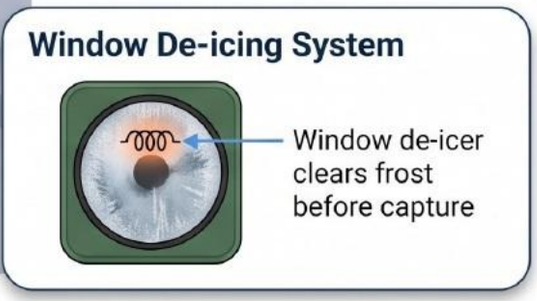
Signed for authenticity, not encrypted

Detection ranges depend on sensor and subject and require validation

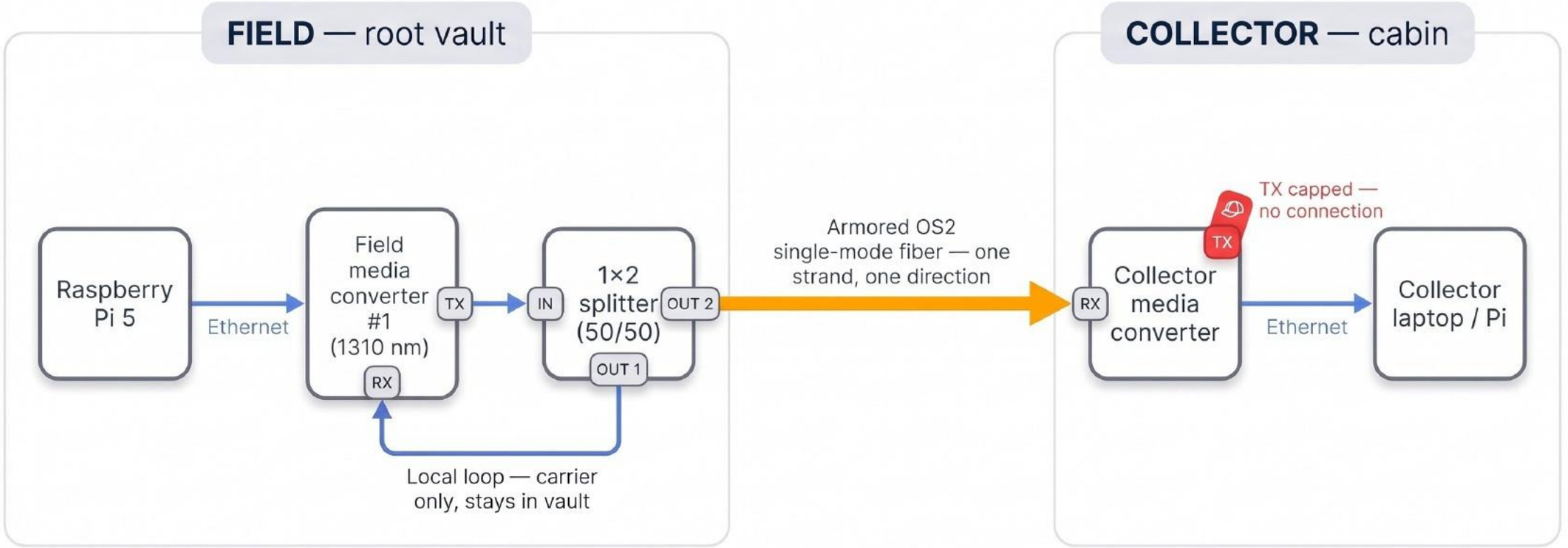
Night Imaging: Invisible 940 nm Light and a Baffle Against Window Glare





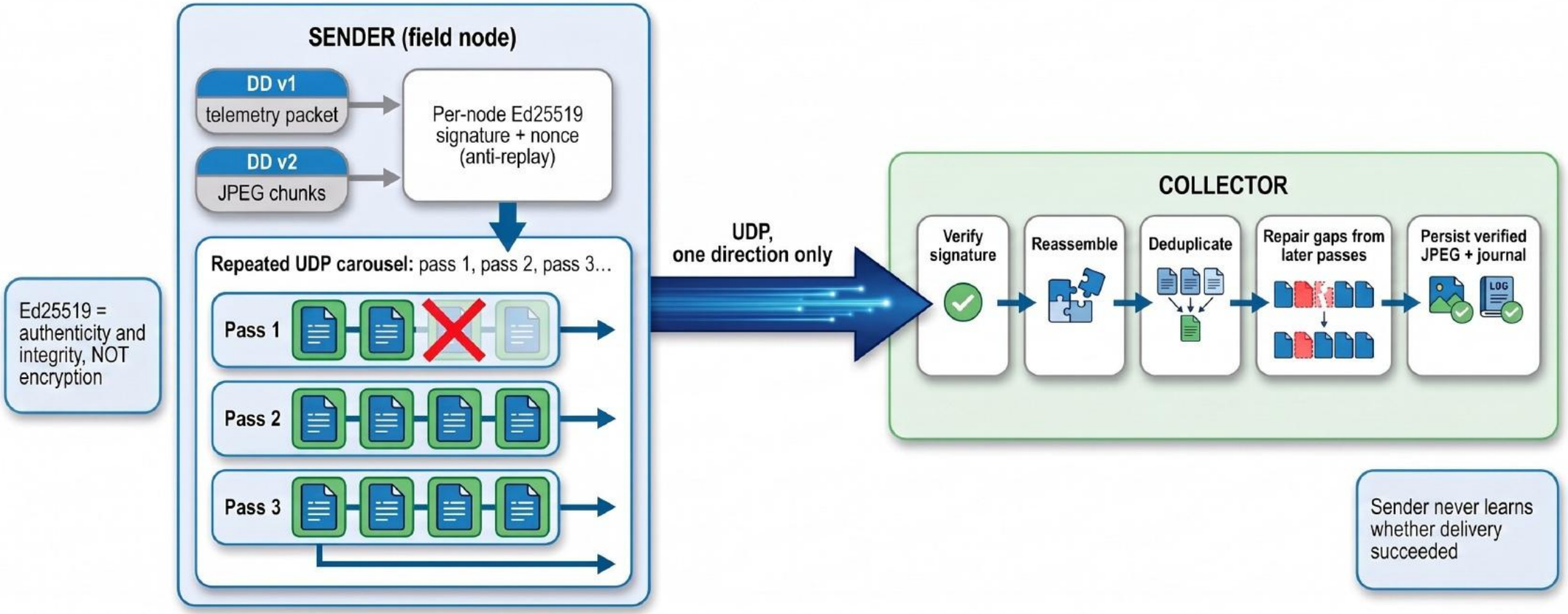


Optical Diode: The Local Loop Keeps the Link Up; Only One Strand Leaves the Tree



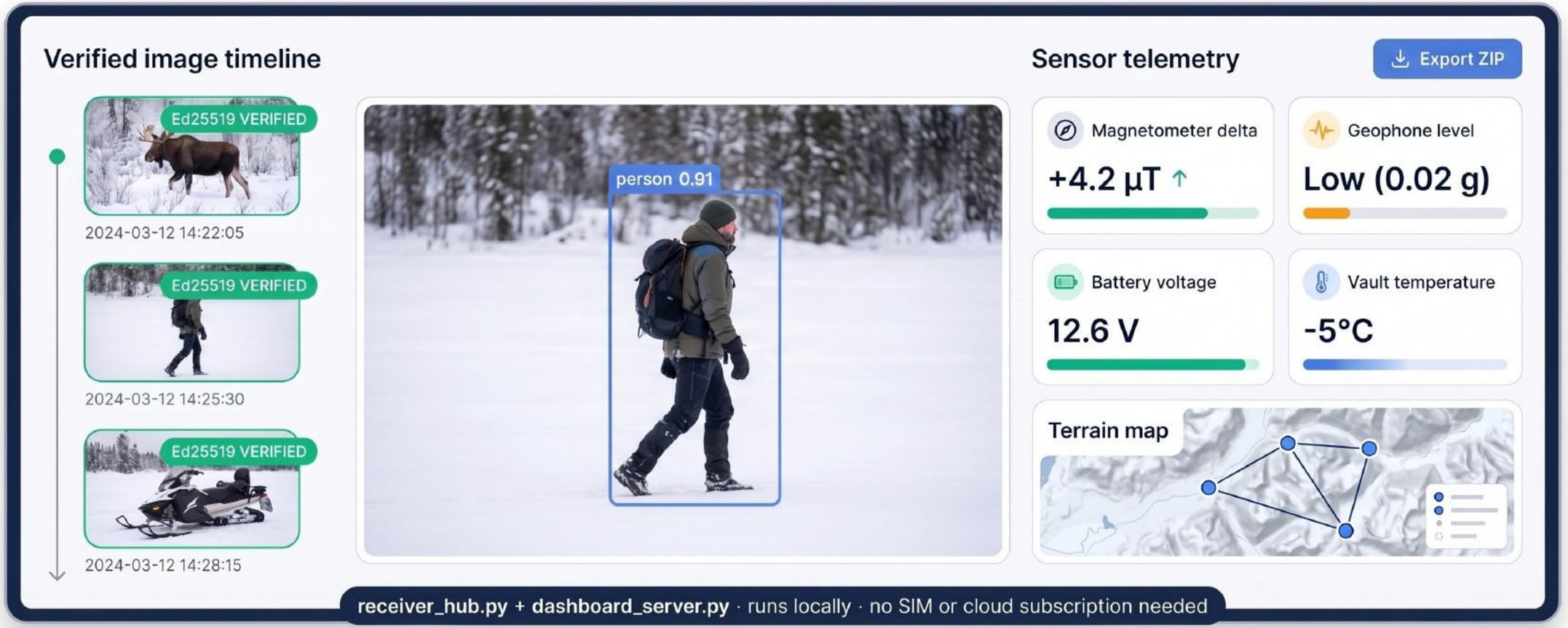
No return command path · No remote ACK · One-way transport does not encrypt or guarantee delivery

Signed Message Journey: Repetition Replaces Acknowledgement



Collector Experience: An Offline Browser Console at the Cabin

Conceptual illustration — not a screenshot

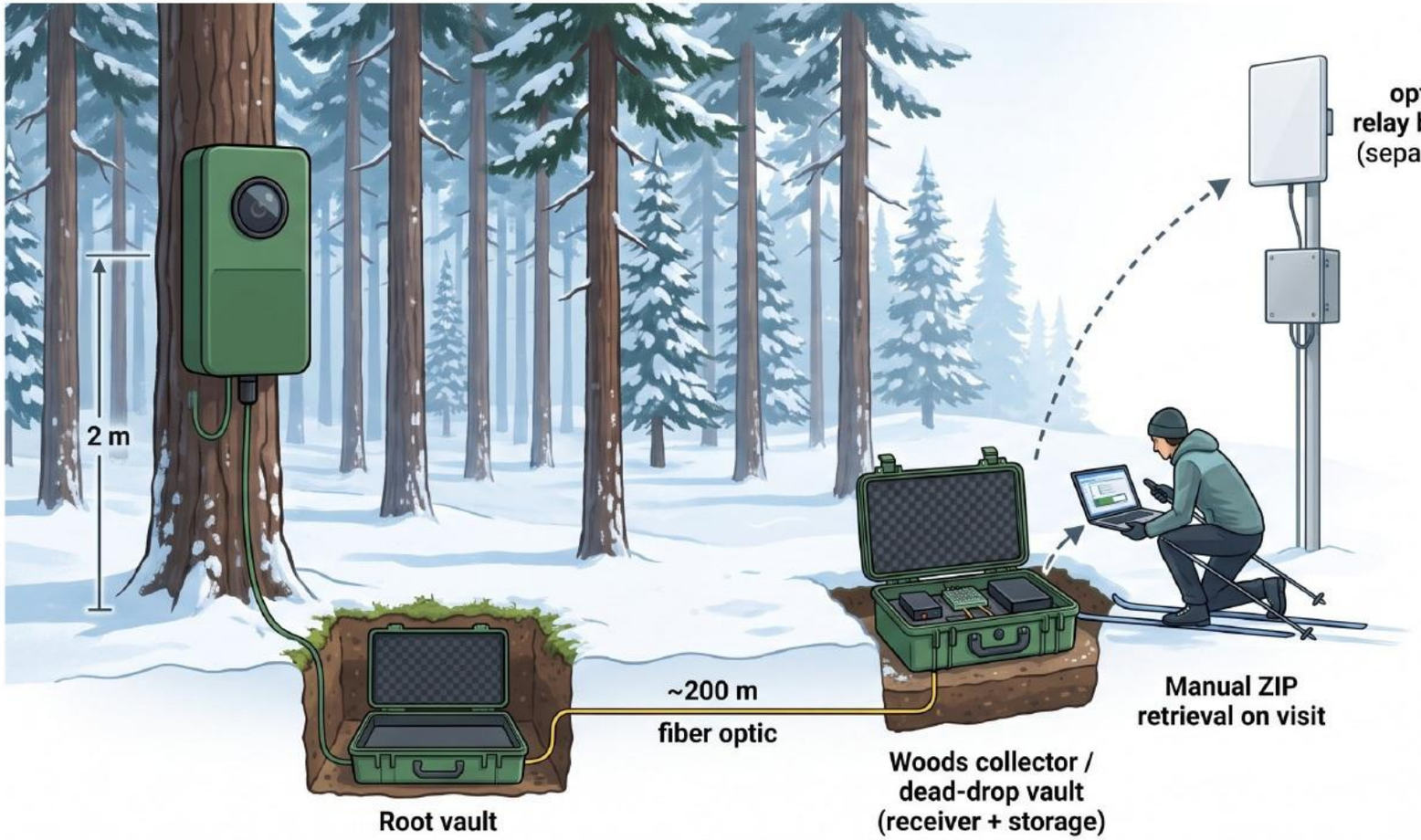


Configuration 1 — Direct Fiber: The Simplest Path When the Cabin Is Within Reach



- ⊕ No RF on the link
- ⊕ Real-time delivery to cabin
- ⊖ Cable must be laid and protected (rodents, frost heave, forestry machines)

Configuration 2 — Woods Dead-Drop: Images Survive Even If the Pod Does Not



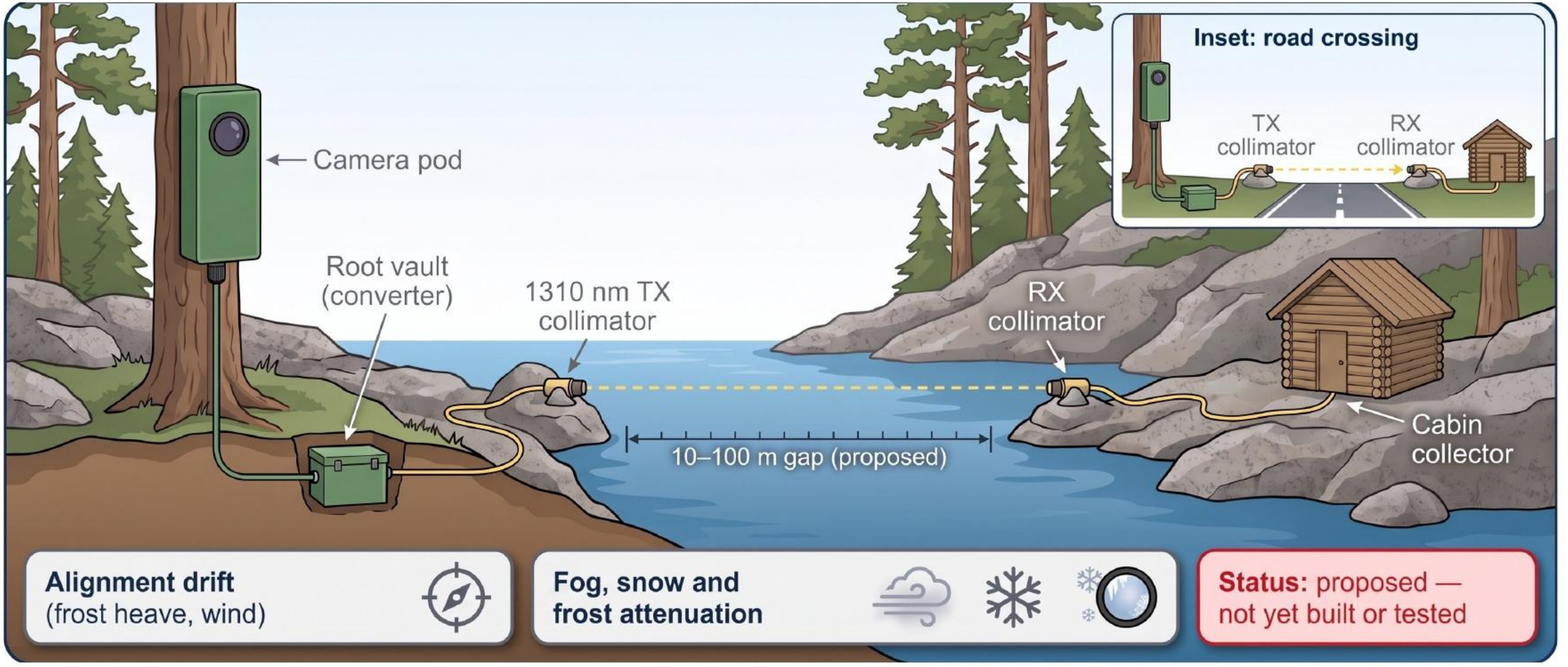
No cabin needed

No live alerts without relay

Photos stored away from the camera

Requires site visits

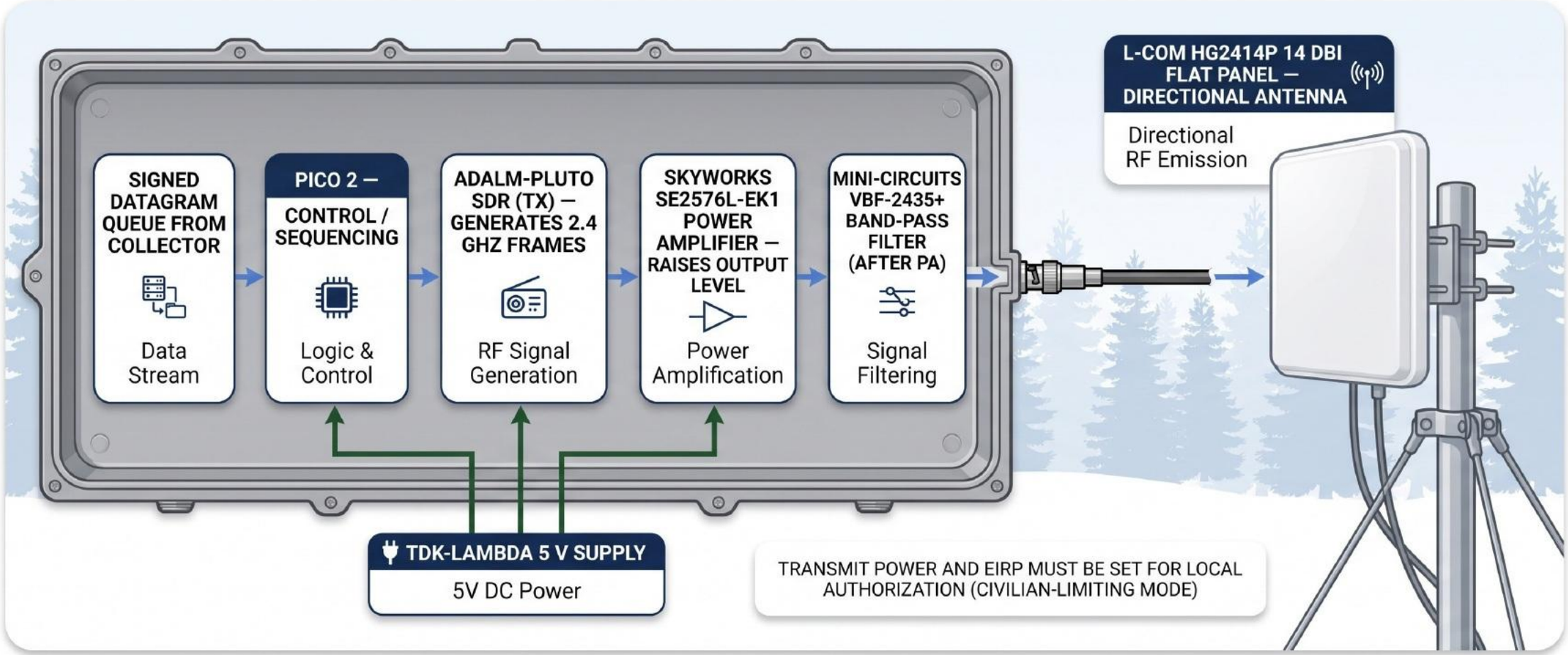
Configuration 3 — Optical Water Bridge (Proposed): Crossing a Gap Without Cable



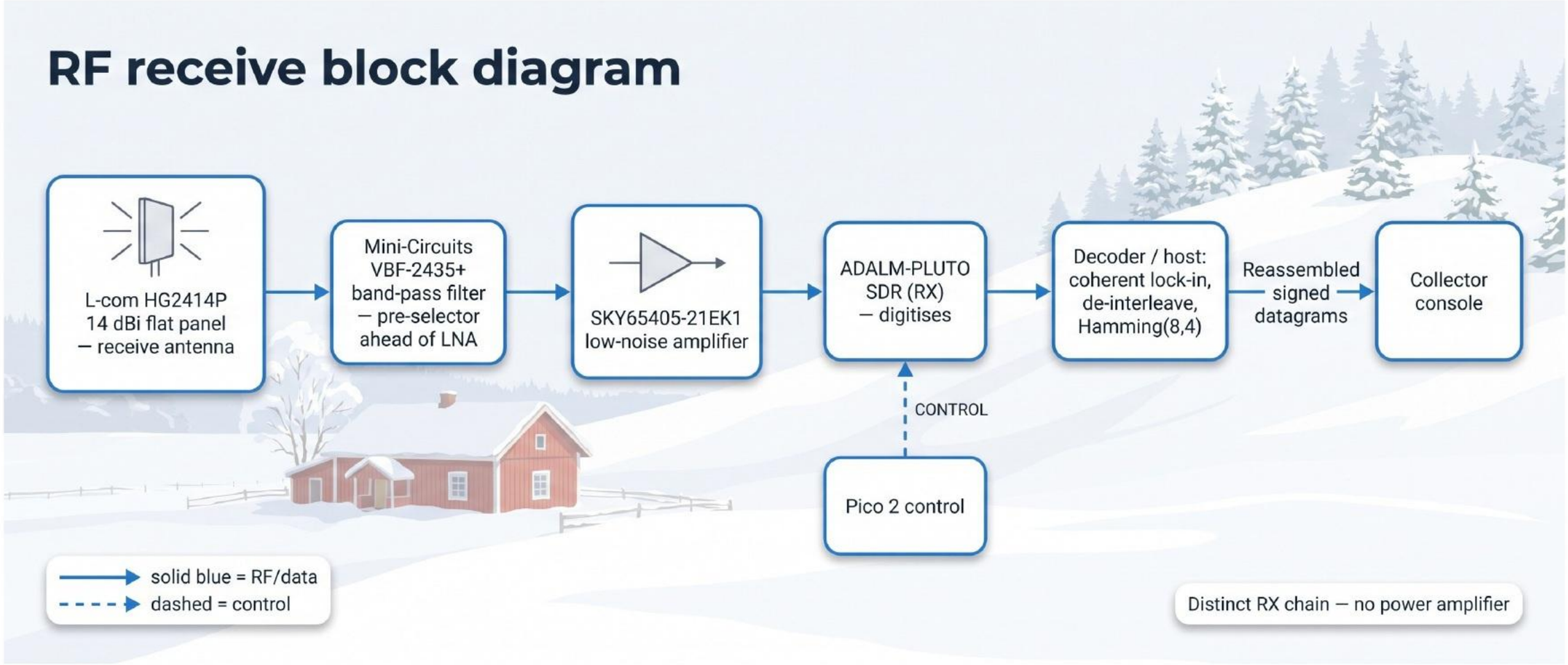
Relay Choices: Match the Path to the Gap

Option	Distance basis	RF emitted?	Main constraint	Status
Direct OS2 fiber	30 m–10 km design envelope	No (optical)	Cable laying and protection	Bench diode reported
Woods dead-drop	~200 m fiber standoff	No (optical)	Manual retrieval	Design
Optical water bridge	10–100 m gap	No (1310 nm light)	Alignment, weather	Proposed
Build D radio relay	200 m–20 km research scope	Yes (2.4 GHz)	Authorization, throughput	Hardware ordered; trials pending
868/915 MHz backhaul	Not quantified	Yes	Authorization, low data rate	Optional
Optical Builds A/B/C	Research alternatives	No	Research-stage	Out of scope here

Build D Transmitter: SDR Signal Shaped, Amplified, Filtered and Aimed

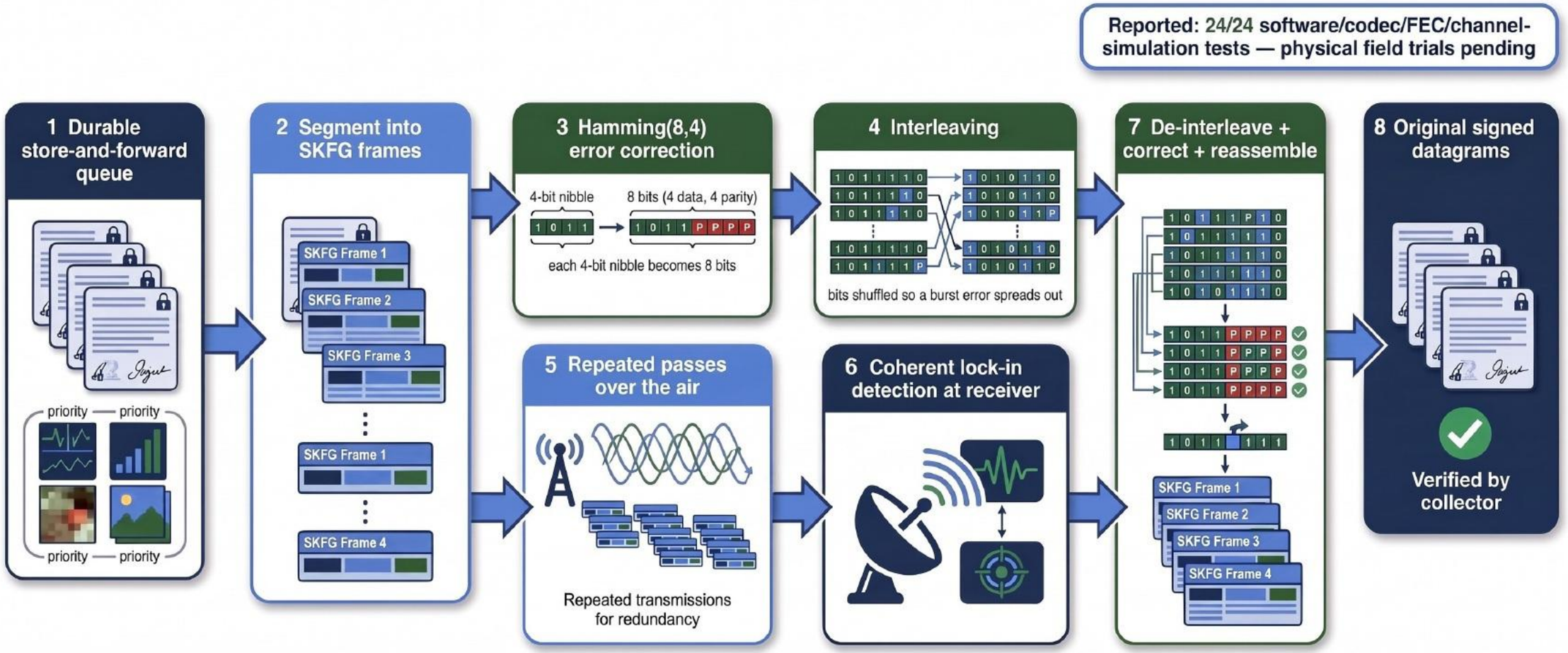


Build D Receiver: Filter First, Then Amplify the Faint Scattered Signal



Power: regulated 5 V to Pico 2, LNA and SDR; host supply as required. Band-pass filter is passive.

Store-and-Forward with Error Correction: Small, Robust Pieces, Sent Repeatedly



Range and Throughput Limits: What the Relay Can and Cannot Claim Today

Reported scope

- 200 m–20 km non-line-of-sight canopy/scatter paths are **model and research scope**
- SNR figures in project PDFs are **model outputs**
- 24/24 software/codec/FEC/channel-simulation tests reported; Build D hardware **ordered**
- Physical relay field trials: **pending**

Not claimed

- No live video and no instantaneous full-image delivery at 20 km
- Throughput and delivery latency **need measurement**
- Delivery is not guaranteed; prioritise small messages and crops
- Weather and season effects on the path are unmeasured

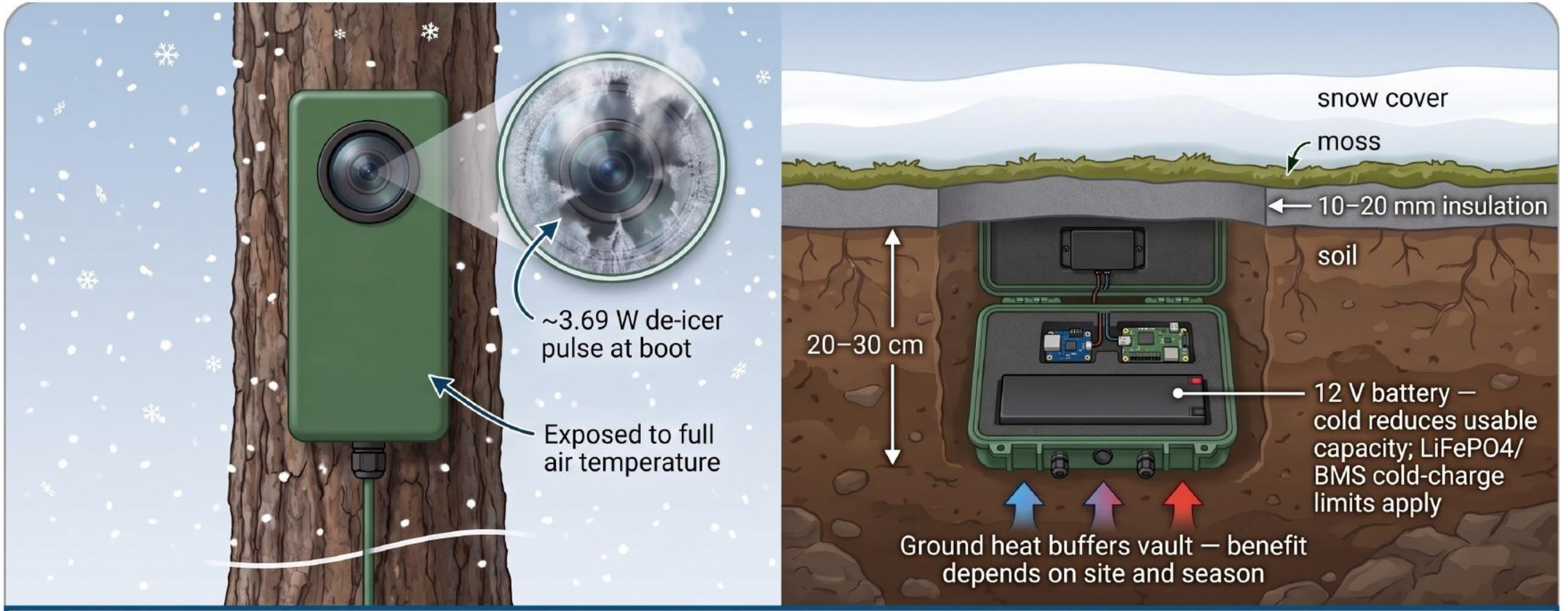
🔗 **Relay sizing has to wait for throughput logs from longer test paths**

Emissions Boundary: Not a Zero-RF System

Element	Emits	Notes
Optical data link (fiber / collimator)	1310 nm light, no RF	Light stays inside the fiber, except across the proposed open-air bridge
Magnetometer, geophone	Nothing intentional	Passive sensing
24 GHz radar (if enabled)	RF	Optional; can be disabled
Build D 2.4 GHz relay	RF	Optional; authorization required
868/915 MHz backhaul	RF	Optional
940 nm IR LEDs	Infrared light	Visible to IR-sensitive cameras
Pi 5 Wi-Fi / Bluetooth	RF if left on	Disable for a no-intentional-RF profile

Passive sensing + optical transport, with Wi-Fi/BT off, avoids *intentional* RF. The system is not zero-RF, undetectable, jam-proof or guaranteed all-weather.

Arctic Thermal Design: Heat the Window Briefly, Keep the Battery Below the Snow



Thermal figures are design targets, not field-measured

Battery Life Projections by Capture Rate and Season

Projected battery life (days) — model projections, not field-measured

Scenario	Lower (days)	Upper (days)
AGM 12 Ah · 10 photos/month · three-season	365	480
AGM 12 Ah · 10 photos/month · winter	270	365
AGM 12 Ah · 10 photos/day	120	220
LiFePO4 15–24 Ah · 10 photos/month	565	900+

All values are model projections, not field measurements. LiFePO4 "900+" denotes an open-ended upper projection of at least 900 days.

Assumptions behind the projections

- Standby draw (3.8–12 mW Eco target)
- Capture rate (~0.025 Wh per capture, modelled)
- Usable capacity and cold derating
- Self-discharge over months
- High-traffic LiFePO4 runtime depends on winter recharge profile
- Battery/BMS cold-charge limits apply

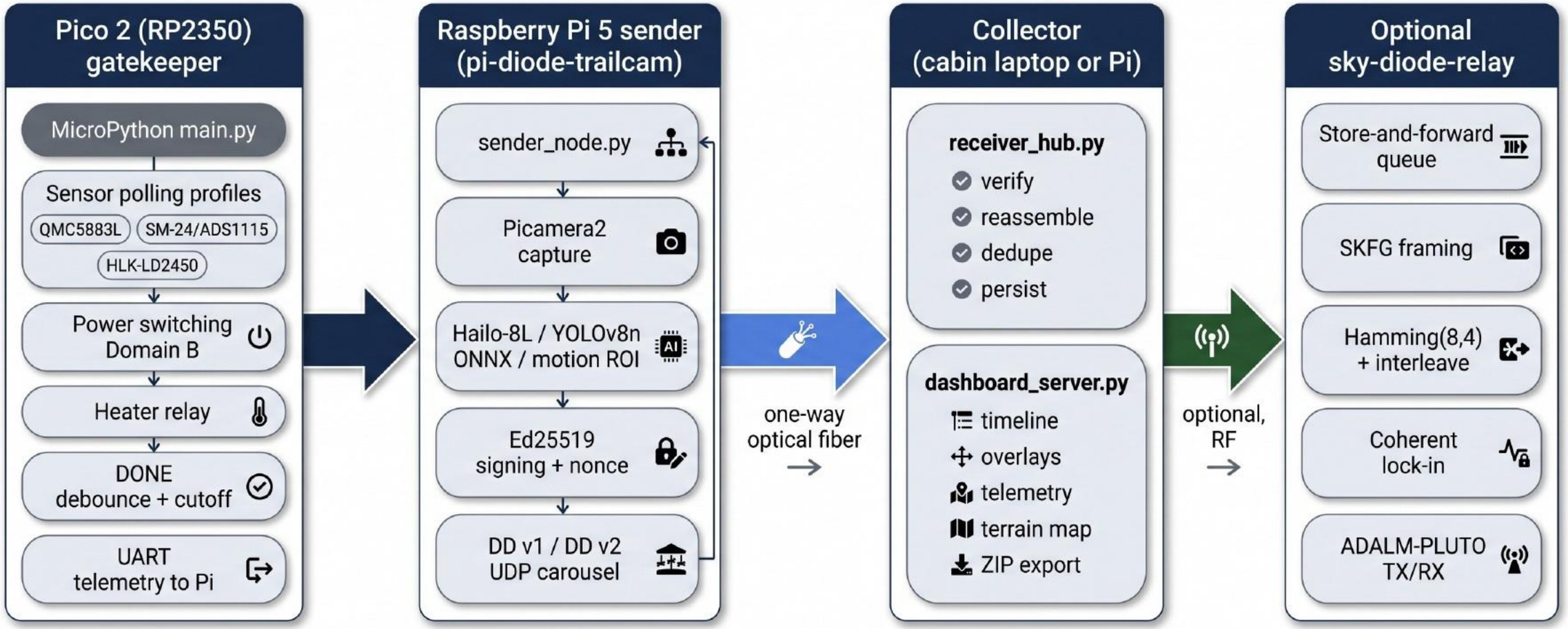
BOM Tiers: A Field Node Plus Fiber Costs About $600; the Radio Relay Is a Separate $1,342

Project BOM snapshot (US$) — not a current quote

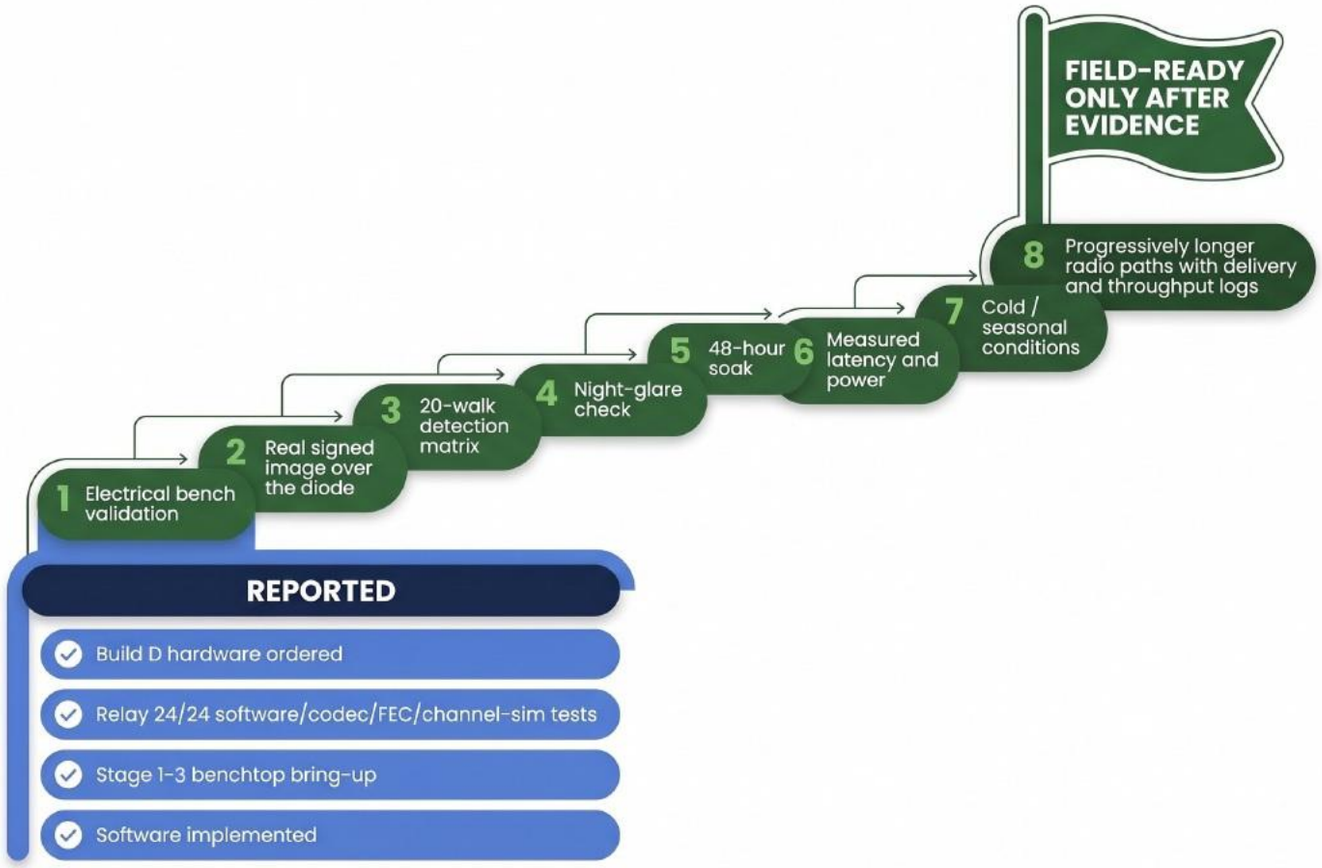
Tier	US$
Core bench	233.37
Bench + diode	402.67
Weatherproof field node only	497.25
Camera + 30 m fiber + cabin RX + cables/adapters	598.20
Build D relay TX+RX pair (separate, not included above)	1,341.94

- **$598.20** covers camera + 30 m fiber + cabin RX + listed cables and adapters. It **excludes** the relay
- **$1,341.94** is the Build D TX+RX parts subtotal, **separate**
- Labor, shipping, taxes, host equipment and optional items may come on top
- Snapshot prices, not latest quotes

Software Map: Who Runs What, Where



Validation Roadmap: Bench Work Reported; Field Evidence Still to Come



Reported bench build (project photo)


[STAGE 2.4C–2.6 BUILD] Real-World Build Log: J-Hook Pin Check, 3 vs 2 Empty Bottom Holes & Powered System (Photos 54–

Chronological Photos 54–58 (1 Step Per Row): Pliers straightening, 3 empty bottom in Col f vs 2 in Col j & live bench system.

ARCTIC TRAIL CAM SYSTEM — STAGE 2.4C–2.6: PIN CHECK, HOLE-COUNT VERIFICATION & POWERED GOLDEN BUILD

FULL-PAGE REAL-WORLD PHOTO BUILD SHEET (1 STEP PER ROW) | ARCTIC TRAIL CAM — REAL-PHOTO BUILD & FIELD MANUAL | Page 9

Stage 2 powered bench system, Build & Field Guide p.9